

16S Report Stub

Overview

Spinner Dolphins were found stranded around Hawaii. Those that have been deceased had a sample of their fecal matter taken. One of the reasons for the stranded dolphins could be the dolphin's gut microbiome. The main question is: Do the gut bacterial communities play a role in spinner dolphins being stranded? Is there a pattern in the metadata that also suggest this?

Data preprocessing

This report analyzes a set of 16S rRNA sequencing data obtained from fecal samples collected from deceased spinner dolphins found stranded around Hawaii. The study aims to investigate the potential role of gut microbiome in the strandings by characterizing the composition and diversity of the spinner dolphin gut microbiome. DNA was extracted from fecal material using the Macherey-Nagel Stool Kit, following the manufacturer's protocol. The 16S rRNA gene was amplified using universal V1–V9 primers (27F + 1492R). PCR amplification and sample barcoding were simultaneously performed using the Oxford Nanopore SQK-16S024 library preparation kit. DNA concentration was quantified using a Qubit Fluorometer with the Qubit High Sensitivity Assay Kit. Barcoded DNA samples were normalized based on concentration and pooled in equimolar amounts. The pooled library was sequenced across multiple Oxford Nanopore Flongle flowcells (FLO-FLG001) for a total sequencing time of 72 hours. Basecalling was performed using the GUPPY v3.2.2 high-accuracy model. Adapter and barcode trimming were conducted during the basecalling process. Taxonomic classification of the reads was carried out using the EPI2ME 16S workflow (Oxford Nanopore Technologies), which employs reference-based alignment for microbial identification. The resulting classified datasets served as the starting point for downstream microbiome analysis.

Loading in the data

First, we will load the required packages and classification data.

```
library(phyloseq)
library(ggplot2)
library(dplyr)
library(tidyr)
library(purrr)
library(vegan)
library(microbiome)

# We will be primarily using the phyloseq and microbiome packages for this analysis
# Documentation can be found by using ?phyloseq and ?microbiome
setwd("/workspaces/r-codespace-rmarkdown/16s")

# Load in the metadata
meta <- read.csv("./metadata.csv") # Add the path to the metadata file

# Load the cleaned data post decontamination
```

```
dat <- read.csv("./kraken.csv")
# Add the path to the cleaned data file
```

Cleaning the data

Next, the feature table obtained after decontamination was cleaned and divided into two components: an OTU table and a taxonomy table. The OTU table contains the barcodes(number of DNA sequences) and the tax table links the OTU to its taxonomic classification. Both tables have different data for different analysis.

```
# Get you bearings on the data
# What functions can you use to get a sense of the data?
names(meta)
str(meta)
names(dat)
str(dat)
# Your bearings on both the metadata and the classification data
```

```
# Check the names of the columns to see where the count/otu data ends and the
# taxonomy information starts
# Clean up and split dat into otu table
otu_dat <- dat[, 2:19] # Add the column number where the count data starts and ends
otu_dat <- otu_table(otu_dat, taxa_are_rows = TRUE)
otu_dat <- t(otu_dat)
# ?otu_table
# Clean up and split dat into taxonomy table
tax_dat <- dat[, c(21:26, 1)]
tax_dat <- tax_table(tax_dat)
# Add the column numbers where the taxonomy data starts and ends

# Add the appropriate function to create the taxonomy table
```

Now that the OTU and taxonomy tables are created, we must make sure the names are consistent between the two tables. This makes sure that it is easy to read each table.

```
# Make sure the taxa_names are the same for otu_dat and tax_dat
# use ?rownames to see how to set the rownames of a dataframe
taxa_names(otu_dat) <- rownames(tax_dat)
# Make sure the rownames are the same for otu_dat and meta
rownames(meta) <- meta$barcode
```

Lastly, a phyloseq object was constructed to integrate the OTU, taxonomy, and metadata information. We do this so that everything is linked by sample and OTU barcode. This is important for graphing and analysis later on.

```
# Create a phyloseq object
# Use ?phyloseq to see how to create a phyloseq object
# Add the appropriate function to create the phyloseq object
ps <- phyloseq(otu_dat, tax_dat, sample_data(meta))

# Name the columns of ps@tax_table
```

```
colnames(ps@tax_table) <- c("genus", "superkingdom", "kingdom", "phylum", "class", "order", "family")
# Add the appropriate column names
```

```
# You can access the phyloseq object by calling ps
head(ps@otu_table)
head(ps@tax_table)
```

Exploratory analysis: Rarefaction curve

In order to see if the sequencing depth is sufficient, we will first create a rarefaction curve of the data. This will show how the number of observed OTUs changes with increasing sequencing depth. This ensures that we collect a sufficient amount unique samples so that the total diversity is reasonable. We conduct the rarefaction analysis by sampling the taxa at different sequencing depths and plotting the number of observed OTUs. In this case we will sample the data at intervals of 100 reads.

```
# Remove taxa with less than 10 reads
ps.rare <- prune_taxa(taxa_sums(ps) > 10, ps) #do this so that we look at data with 10 reserach points

# Add the appropriate function to prune taxa with less than 10 reads

# Create a rarefaction curve

# create an OTU table based on the pruned data
rare_table <- otu_table(ps.rare)

#convert the OTU table to a dataframe
rare_table <- as.data.frame(otu_table(ps.rare))
sample_names <- rownames(rare_table)

# use the rownames of the rare_table to create the sample names

raredat <- rarecurve(rare_table, step = 100, label = TRUE, tidy = TRUE)

# Add the appropriate function to create the rarefaction curve; use ?rarecurve for help

rareplot <- ggplot(raredat, aes(x = Sample, y = Species, color = Site)) +
  geom_point()

# Plot the rarefaction curve
# Use ?ggplot to see how to create a ggplot object

# Display the rarefaction curve
rareplot
```

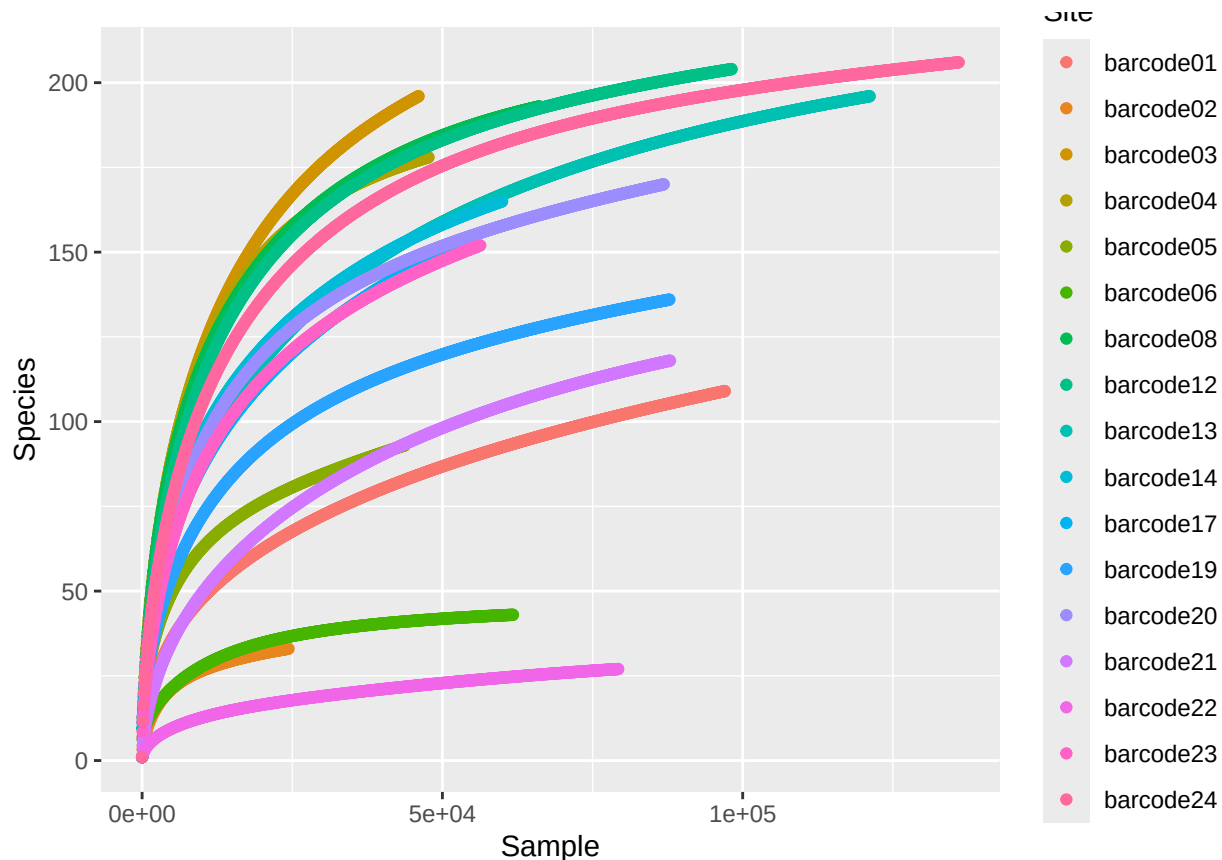


Figure 1: Rarefaction curve of the Spinner Dolphins microbiome.

The rarefaction curve shows that the sequencing depth is enough for majority of the samples. As the number of barcodes from the OTU begin to plateau, that means that most of the unique species have been gathered, making total diversity as accurate as possible. ## Exploratory analysis: Relative abundance

We will now conduct an exploratory analysis of the data to determine the relative abundance of the bacterial taxa in the samples. To clean up the data, we will first remove any taxa contain <100 reads since they are regarded as “noise.” This can skew data and make patterns harder to detect. We will first calculate the relative abundance of the taxa in the samples. Since different samples have different sequencing depths. Calculating the relative abundance will allow us to compare microbiome composition proportionally. We will then plot the relative abundance of the taxa in the samples to visually see the abundance of species in the Spinner Dolphins.

```
# Remove taxa with less than 100 reads3
#ps.prune <- # Add the appropriate function to prune taxa with less than 100 reads

ps.prune <- prune_taxa(taxa_sums(ps) > 100, ps)

# Consolidate the taxa to the family level
cfam <- tax_glom(ps.prune, taxrank = rank_names(ps.prune)[5])

# use tax_glom to consolidate the taxa to the family level

# Calculate the relative abundance
cfam <- transform_sample_counts(cfam, function(x) x / sum(x) * 100 )
# use transform_sample_counts to calculate the relative abundance
```

```
melt <- psmelt(cfam)
melt <- melt %>%
  group_by(id, Location, class) %>%
  summarize(RelativeAbundance = sum(Abundance))
```

'summarise()' has grouped output by 'id', 'Location'. You can override using the '.groups' argument.

```
# Subset the samples by metadata of interest
```

```
# Plot relative abundance BASED ON METADATA OF INTEREST
# use ?plot_bar to see how to create a bar plot
plot <- ggplot(melt, aes(x = id, y = RelativeAbundance, fill = class)) +
  geom_bar(stat = 'identity') +
  ggtitle("Relative Abundance of Spinner Dolphin species")

# use ?geom_bar to see how to create a bar plot
```

```
# Plot relative abundance BASED ON METADATA OF INTEREST
# use ?plot_bar to see how to create a bar plot
# use ?geom_bar to see how to create a bar plot
plot
```

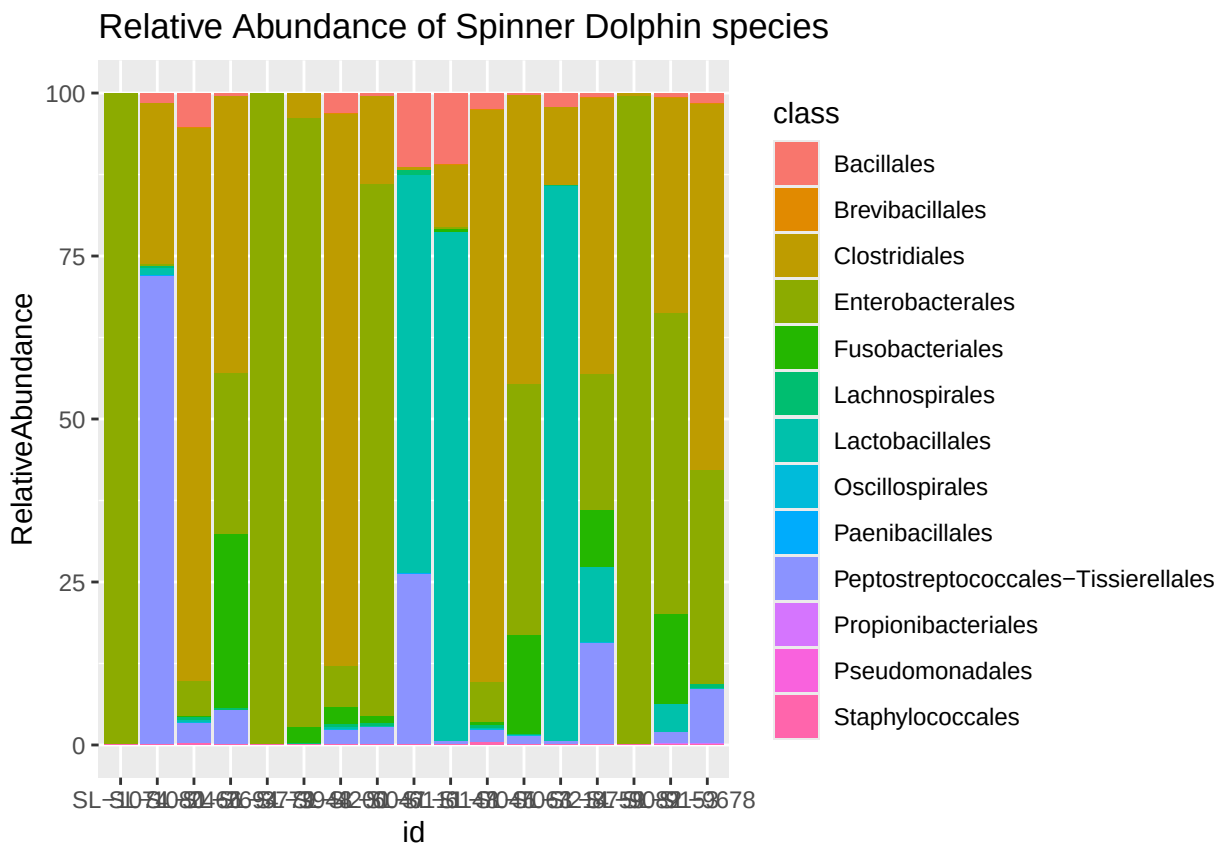


Figure 1: Relative abundance of FILL THIS OUT FOR YOUR DATASET

Figure 2: Relative abundance of the bacterial orders in the Spinner Dolphins

From looking at this relative abundance barplot, we can see that a majority of the Spinner Dolphins have a high abundance of microbiome consisting of Clostridiales(orange) and Enterobacterales(green). While there are differences across the Spinner Dolphins microbiome makeup, we will need to conduct a more rigorous analysis to determine if the samples are actually different from each other. *## Alpha diversity analysis*

In order to quantify the diversity within the samples, we will calculate the alpha diversity of the samples. We will calculate the observed species and Simpson diversity indices for the samples. The observed species index measures the number of unique species present in the samples, while the Simpson index indicates the diversity of the sample. We will then plot the alpha diversity indices for the samples. A lower richness could indicate that a lower bacterial diversity leads to the dolphins being stranded. With the Simpson index, we can find out if the Dolphins microbiome is dominated some bacterial groups.

```
# Calculate alpha diversity  
# use ?plot_richness to calculate the observed species
```

```
alpha_indiv1 <- plot_richness(ps, x = "samples", measures = "Observed") +  
  geom_boxplot(aes(color = samples))
```

```
## Warning: 'aes_string()' was deprecated in ggplot2 3.0.0.  
## i Please use tidy evaluation idioms with 'aes()'.  
## i See also 'vignette("ggplot2-in-packages")' for more information.  
## i The deprecated feature was likely used in the phyloseq package.  
## Please report the issue at <https://github.com/joey711/phyloseq/issues>.  
## This warning is displayed once every 8 hours.  
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was generated.
```

```
# use ?plot_richness to calculate the Simpson diversity index
```

```
alpha_indiv2<- plot_richness(ps,x = "samples",measures= "Simpson")+  
  geom_boxplot(aes(color = samples))
```

```
alpha_indiv3 <- plot_richness(ps, x = "Age.class", measures = "Observed") +  
  geom_boxplot(aes(color = Age.class))
```

```
# use ?plot_richness to calculate the Simpson diversity index
```

```
alpha_indiv4<- plot_richness(ps,x = "Age.class",measures= "Simpson")+  
  geom_boxplot(aes(color = Age.class))
```

```
alpha_indiv5 <- plot_richness(ps, x = "Sex", measures = "Observed") +  
  geom_boxplot(aes(color = Sex))
```

```
# use ?plot_richness to calculate the Simpson diversity index
```

```
alpha_indiv6 <- plot_richness(ps,x = "Sex",measures= "Simpson")+  
  geom_boxplot(aes(color = Sex))
```

```
# display the plot
```

```
alpha_indiv1
```

```
alpha_indiv3
```

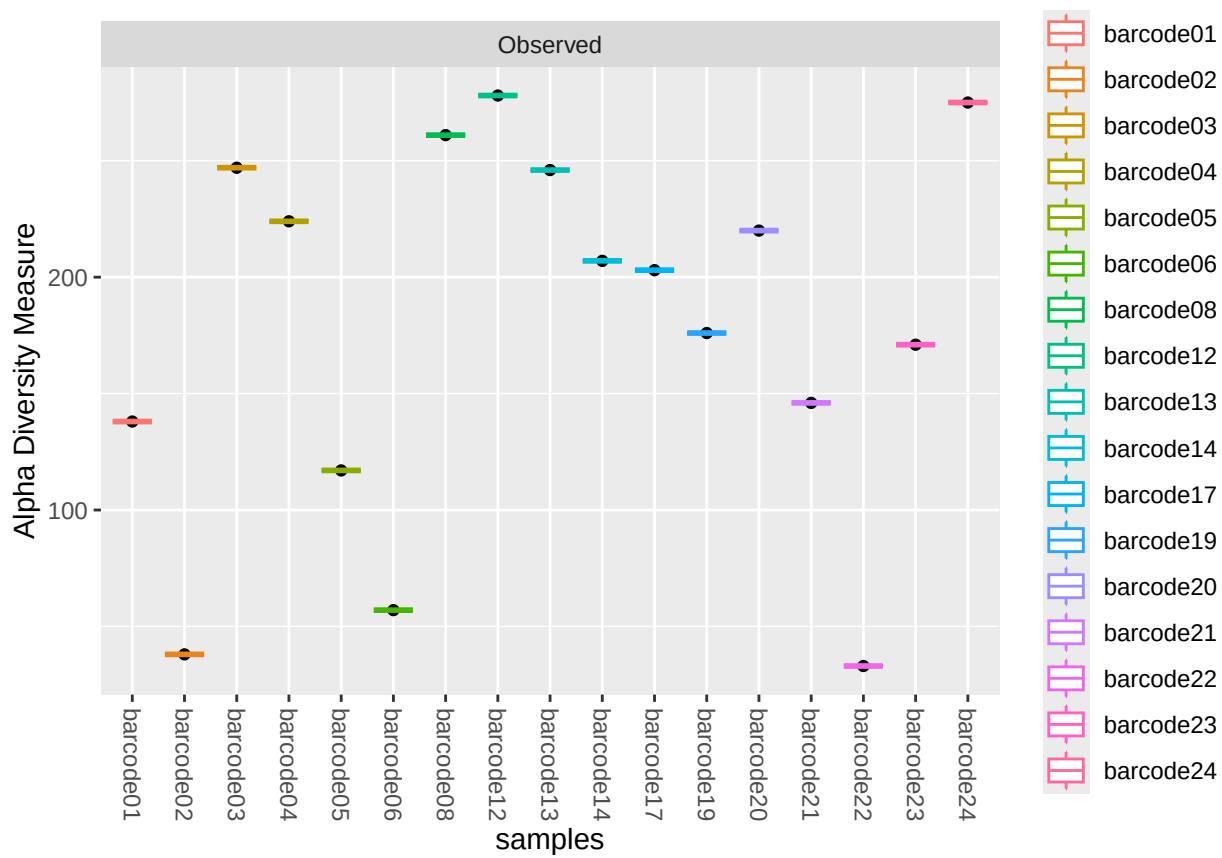


Figure 2: Observed species richness BASED ON YOUR DATASET.

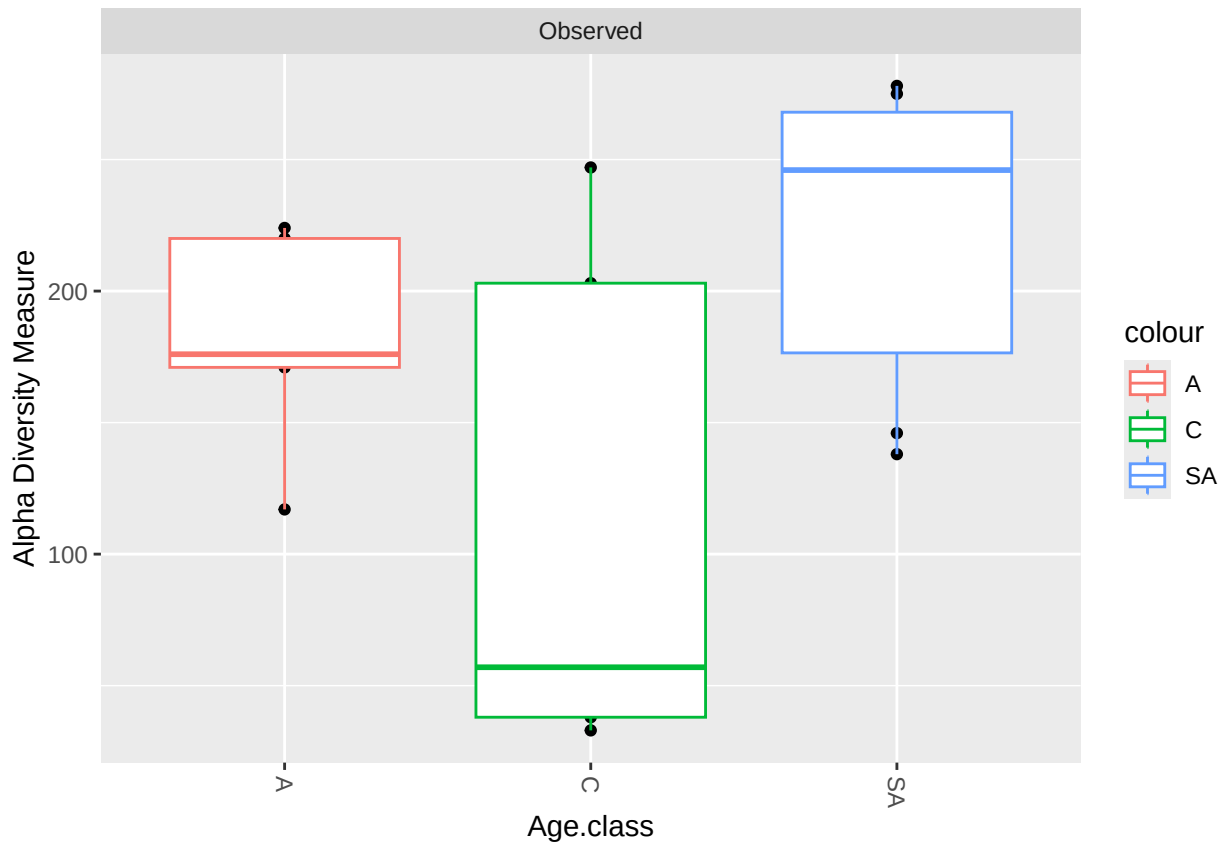


Figure 3: Observed species richness BASED ON YOUR DATASET.

```
alpha_indiv5
```

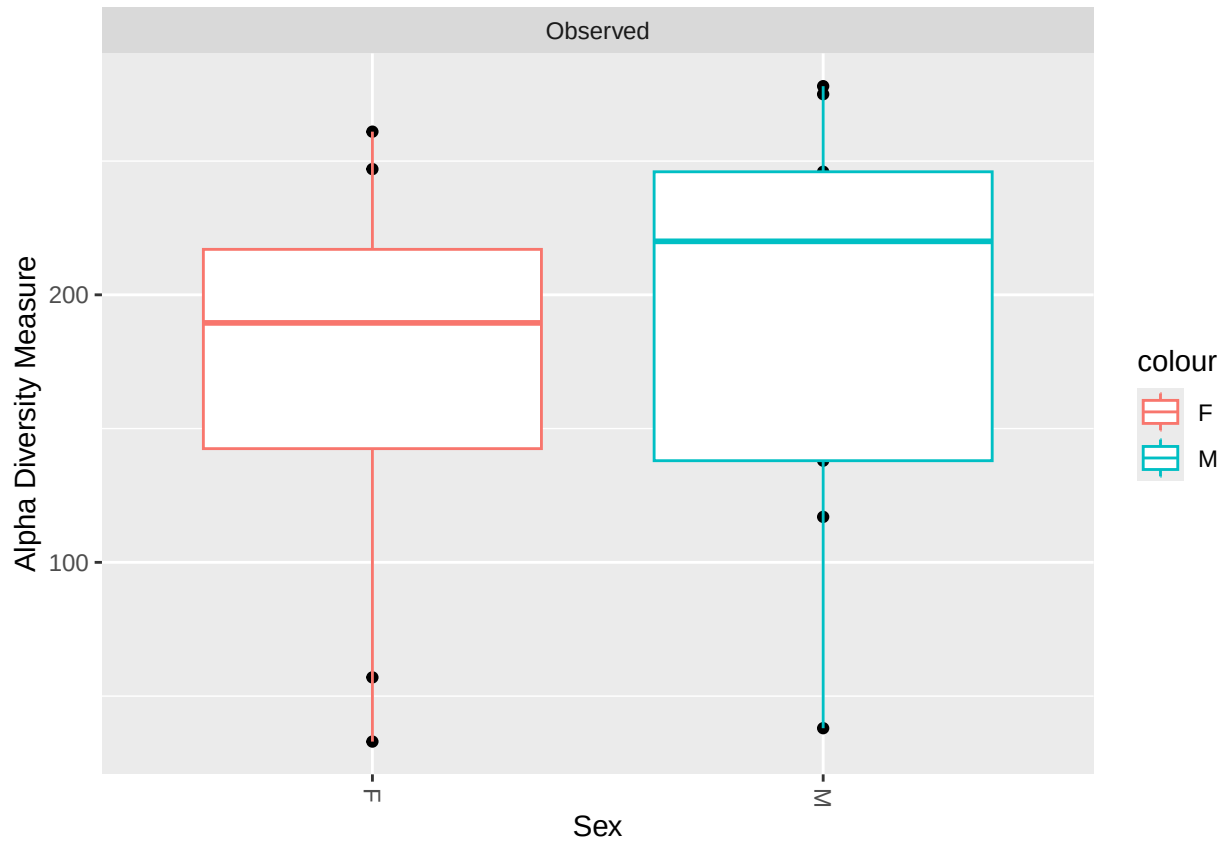


Figure 3: Observed species richness of the Spinner Dolphins microbiome. Figure 4: Observed Age.Class richness of the Spinner Dolphins microbiome. Figure 5: Observed Sex richness of the Spinner Dolphins microbiome.

```
# display the plot  
alpha_indiv2
```

```
alpha_indiv4
```

```
alpha_indiv6
```

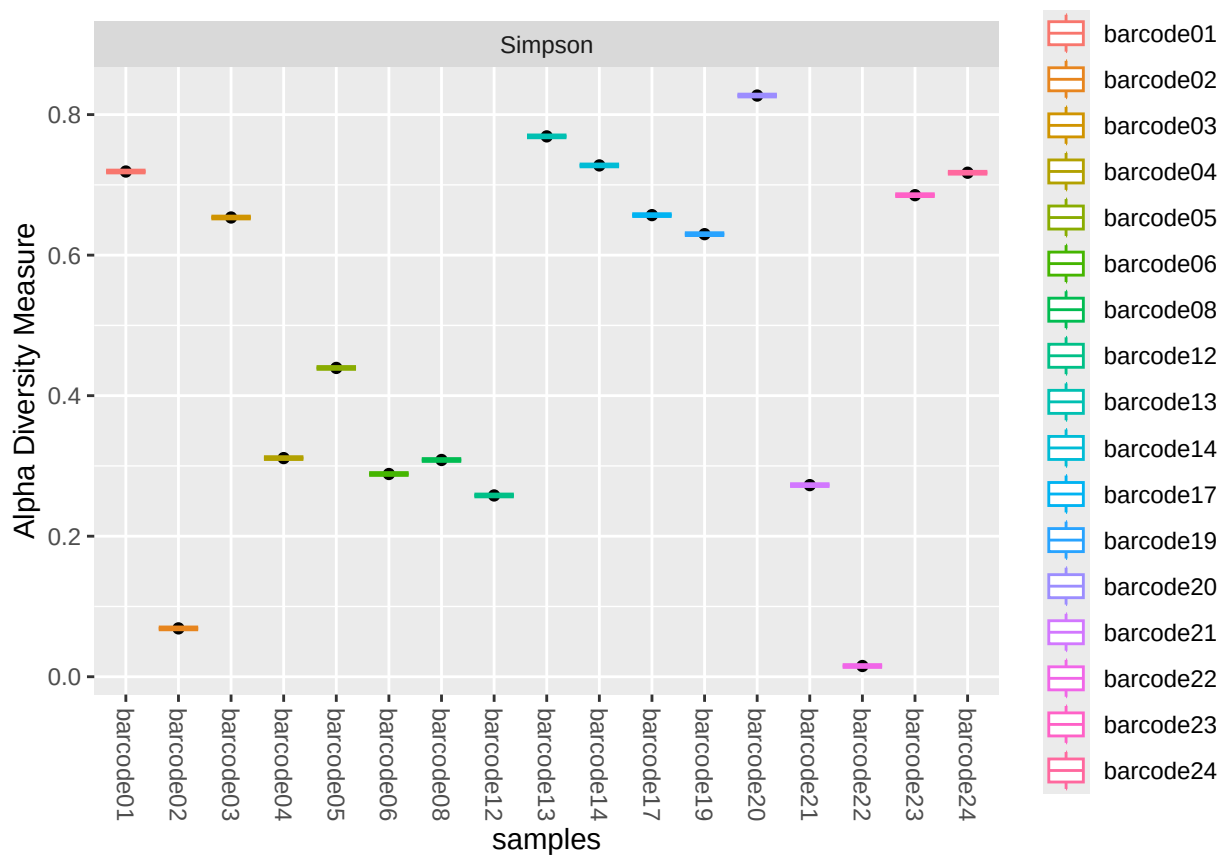


Figure 4: Simpsons diversity index BASED ON YOUR DATASET.

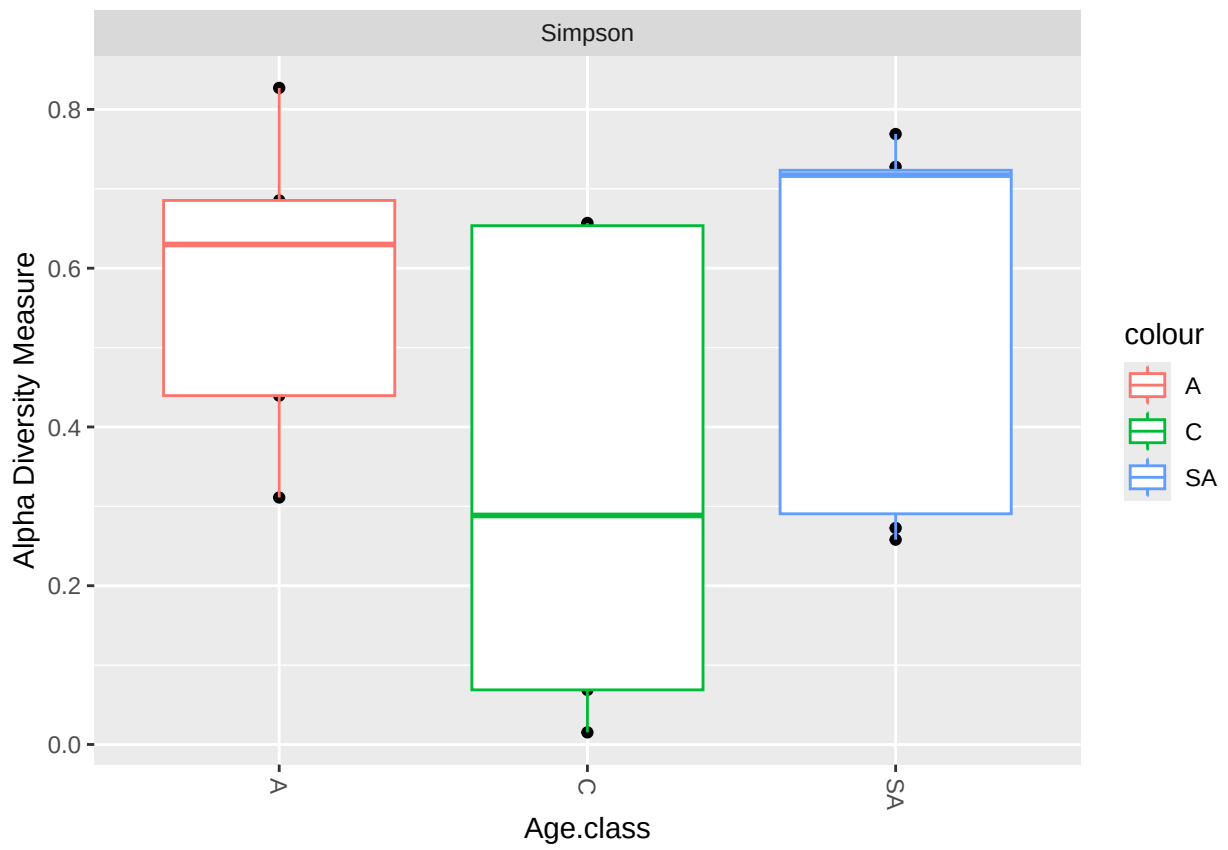


Figure 5: Simpsons diversity index BASED ON YOUR DATASET.

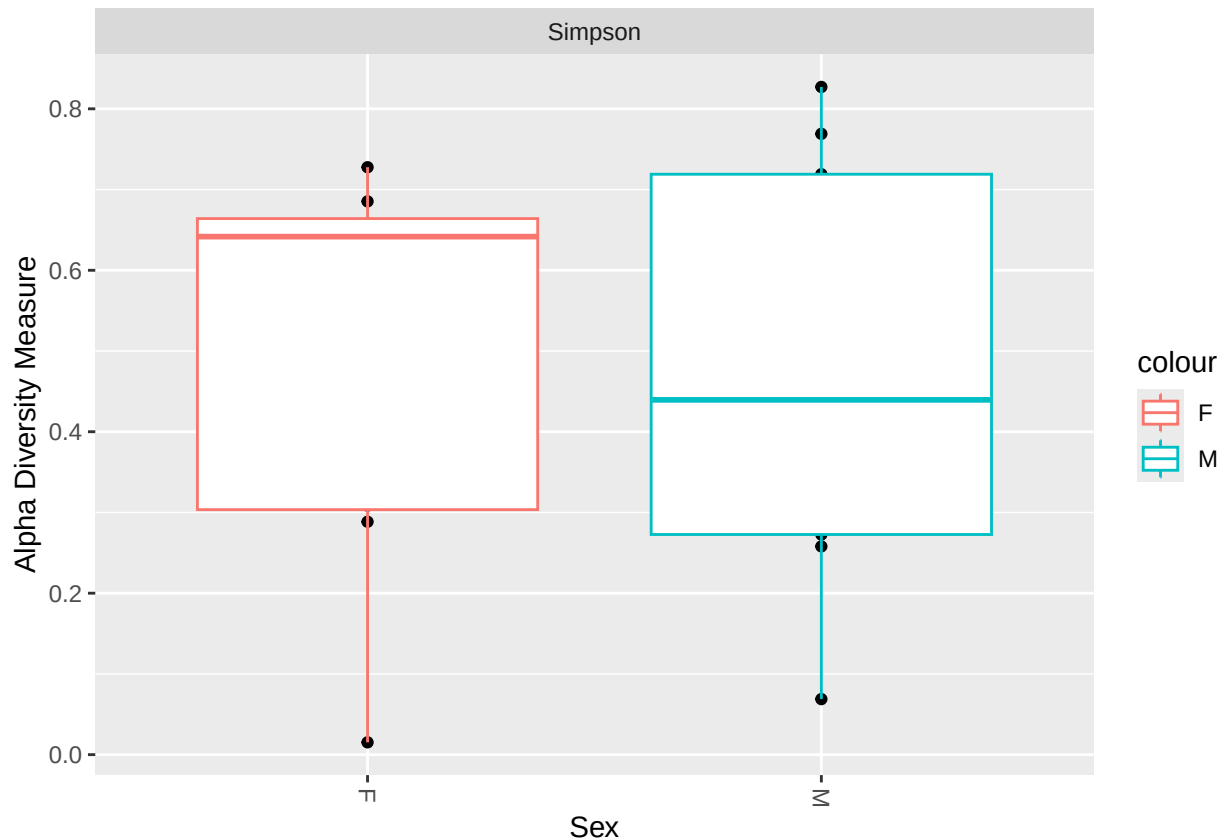


Figure 6: Simpson species richness of the Spinner Dolphins microbiome. Figure 7: Simpson Age.Class richness of the Spinner Dolphins microbiome. Figure 8: Simpson Sex richness of the Spinner Dolphins microbiome.

When looking at the Observed index, a higher index indicates a high amount of diversity. The Simpson index tells us how evenly distributed the sample is. When looking at the observed vs Simpson index of the samples, there is a cluster of Spinner Dolphins with medium/high diversity; medium/high spread. This indicates that the gut bacterial community is not the best indicator for the Spinner Dolphins being stranded. When looking at the sex and age class, they could potentially be indicators for the Dolphins being stranded. With the Males having a lower Simpson index and the Calves having a lower Simpson and Observed index, they are factors that contribute to the Dolphins being stranded. ## Exploratory analysis: PCoA

We will now conduct a principle coordinate analysis (PCoA) of the samples to detect similarities in bacterial composition at the genus/species level. We will first convert the data to compositional data and then create a distance matrix based on Bray-Curtis dissimilarity. The distance matrix will then be used to conduct the PCoA. The results of the PCoA will be plotted.

```
# Create a distance matrix based on bray-curtis dissimilarity
# use ?ordinate to see how to create a distance matrix and PCoA
ord <- ordinate(ps,method="MDS")
```

We will now plot the PCoA results.

```
# Plot the PCoA results and save as an object
# Use ?plot_ordination to see how to create a PCoA plot
pcoa_plot <- plot_ordination(ps,ord,color="barcode")

# Display the plot
pcoa_plot
```

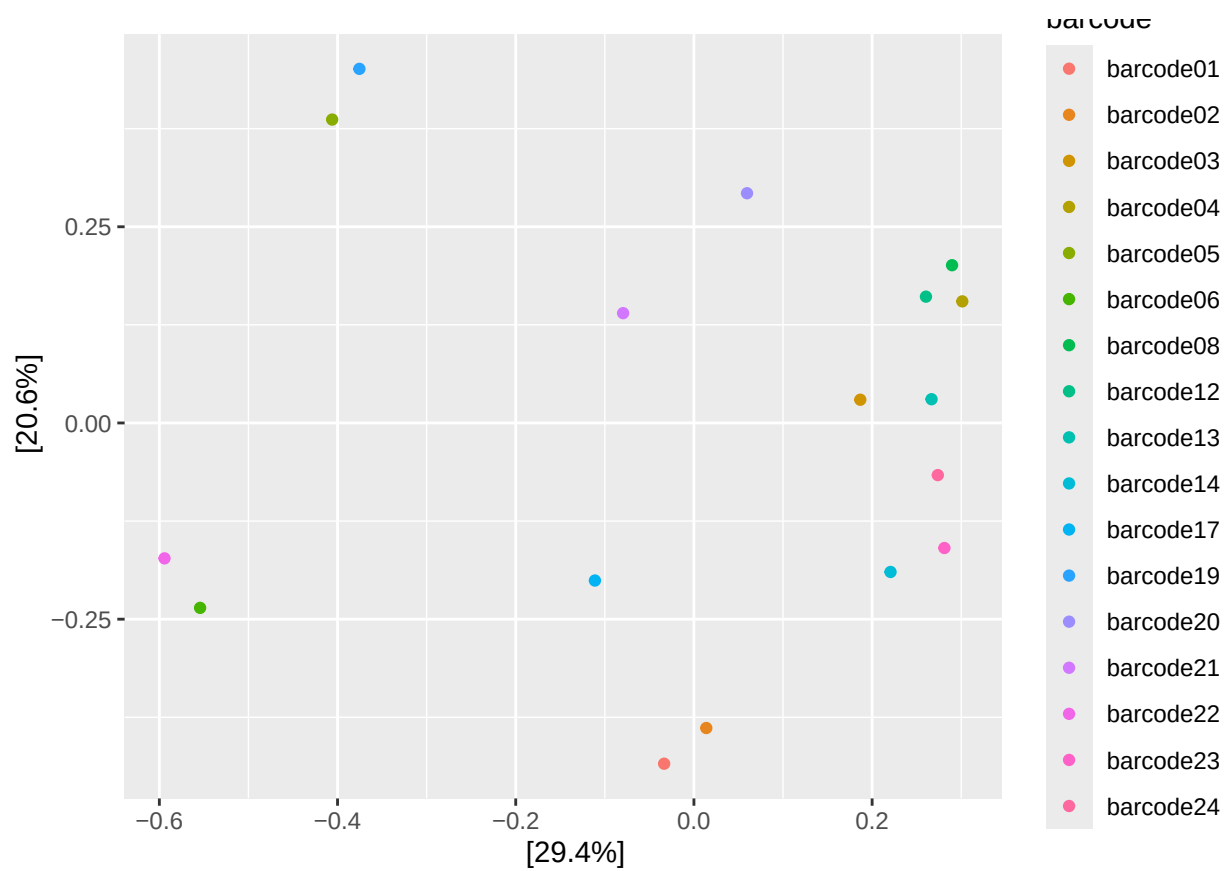


Figure 6: PCoA of the native Hawaiian snail microbiome colored by species.

From the PCoA plot, there is a small cluster on the right while everything else is scattered. This indicates that there are a few samples where Dolphins have similar bacterial makeup. There are also a few Dolphins where there is variation in their bacterial makeup. This means that there is some variability in bacterial composition. ## Conclusions From the analysis, we can conclude that the microbiome for the Spinner Dolphins is not a clear indicative of the Dolphins being stranded. Sampling depth for all samples appears to be sufficient as indicated by the rarefaction curve (Fig. 1). Although the relative abundance of species in the Spinner Dolphin seem similar from the barplot, the PCoA indicates that there is some variability. This means that the microbiomes of the Spinner Dolphin tend to vary individually. However, when looking at some of the metadata of the Spinner Dolphins, Male Calves seem to be a potential indicative for the Dolphins being stranded. This is due to their lower Simpson rates and low Observed rates. This information can be used to inform organizations of the demographic of Dolphins they could put their attention on to help them. Further analysis is required to determine if the microbiomes of Spinner Dolphins are species specific and if it is a reliable indicative of Spinner Dolphins being stranded.

```
library(rmarkdown)
```

```
library(tinytex)
```

```
render("16s_stub_v2.Rmd", output_format = "pdf_document")
```